



Top 41 CAT Selection With Condition Questions With Video Solutions

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Questions

Instructions

Three reviewers Amal, Bimal, and Komal are tasked with selecting questions from a pool of 13 questions (Q01 to Q13). Questions can be created by external “subject matter experts” (SMEs) or by one of the three reviewers. Each of the reviewers either approves or disapproves a question that is shown to them. Their decisions lead to eventual acceptance or rejection of the question in the manner described below.

If a question is created by an SME, it is reviewed first by Amal, and then by Bimal. If both of them approve the question, then the question is accepted and is not reviewed by Komal. If both disapprove the question, it is rejected and is not reviewed by Komal. If one of them approves the question and the other disapproves it, then the question is reviewed by Komal. Then the question is accepted only if she approves it.

A question created by one of the reviewers is decided upon by the other two. If a question is created by Amal, then it is first reviewed by Bimal. If Bimal approves the question, then it is accepted. Otherwise, it is reviewed by Komal. The question is then accepted only if Komal approves it. A similar process is followed for questions created by Bimal, whose questions are first reviewed by Komal, and then by Amal only if Komal disapproves it. Questions created by Komal are first reviewed by Amal, and then, if required, by Bimal.

The following facts are known about the review process after its completion.

1. Q02, Q06, Q09, Q11, and Q12 were rejected and the other questions were accepted.
2. Amal reviewed only Q02, Q03, Q04, Q06, Q08, Q10, Q11, and Q13.
3. Bimal reviewed only Q02, Q04, Q06 through Q09, Q12, and Q13.
4. Komal reviewed only Q01 through Q05, Q07, Q08, Q09, Q11, and Q12.

Question 1

How many questions were DEFINITELY created by Amal?

not reviewed by Komal. If both disapprove the question, it is rejected and is not reviewed by Komal. If one of them approves the question and the other disapproves it, then the question is reviewed by Komal. Then the question is accepted only if she approves it.

A question created by one of the reviewers is decided upon by the other two. If a question is created by Amal, then it is first reviewed by Bimal. If Bimal approves the question, then it is accepted. Otherwise, it is reviewed by Komal. The question is then accepted only if Komal approves it. A similar process is followed for questions created by Bimal, whose questions are first reviewed by Komal and then by Amal only if Komal disapproves it. Questions created by Komal are first reviewed by Amal, and then, if required, by Bimal.

The following facts are known about the review process after its completion.

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Question 2

How many questions were DEFINITELY created by the SMEs?

not reviewed by Komal. If both disapprove the question, it is rejected and is not reviewed by Komal. If one of them approves the question and the other disapproves it, then the question is reviewed by Komal. Then the question is accepted only if she approves it.

A question created by one of the reviewers is decided upon by the other two. If a question is created by Amal, then it is first reviewed by Bimal. If Bimal approves the question, then it is accepted. Otherwise, it is reviewed by Komal. The question is then accepted only if Komal approves it. A similar process is followed for questions created by Bimal, whose questions are first reviewed by Komal, and then by Amal only if Komal disapproves it. Questions created by Komal are first reviewed by Amal, and then, if required, by Bimal.

The following facts are known about the review process after its completion.

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Question 3

How many questions were DEFINITELY created by Komal?

not reviewed by Komal. If both disapprove the question, it is rejected and is not reviewed by Komal. If one of them approves the question and the other disapproves it, then the question is reviewed by Komal. Then the question is accepted only if she approves it.

A question created by one of the reviewers is decided upon by the other two. If a question is created by Amal, then it is first reviewed by Bimal. If Bimal approves the question, then it is accepted. Otherwise, it is reviewed by Komal. The question is then accepted only if Komal approves it. A similar process is followed for questions created by Bimal, whose questions are first reviewed by Komal, and then by Amal only if Komal disapproves it. Questions created by Komal are first reviewed by Amal, and then, if required, by Bimal.

The following facts are known about the review process after its completion.

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Question 4

The approval ratio of a reviewer is the ratio of the number of questions (s)he approved to the number of questions (s)he reviewed. Which option best describes Amal's approval ratio?

- A either 0.25 or 0.75
- B 0.25
- C lies between 0.25 and 0.50
- D lies between 0.25 and 0.75

not reviewed by Komal. If both disapprove the question, it is rejected and is not reviewed by Komal. If one of them approves the question and the other disapproves it, then the question is reviewed by Komal. Then the question is accepted only if she approves it.

A question created by one of the reviewers is decided upon by the other two. If a question is created by Amal, then it is first reviewed by Bimal. If Bimal approves the question, then it is accepted. Otherwise, it is reviewed by Komal. The question is then accepted only if Komal approves it. A similar process is followed for questions created by Bimal, whose questions are first reviewed by Amal, and then by Amal only if Komal disapproves it. Questions created by Komal are first reviewed by Amal, and then, if required, by Bimal.

The following facts are known about the review process after its completion.



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Question 5

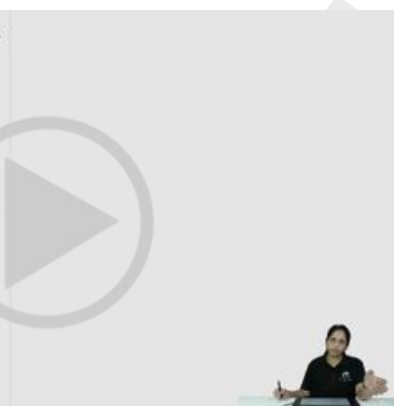
How many questions were DEFINITELY disapproved by Bimal?

- A 3
- B 4
- C 7
- D 5

not reviewed by Komal. If both disapprove the question, it is rejected and is not reviewed by Komal. If one of them approves the question and the other disapproves it, then the question is reviewed by Komal. Then the question is accepted only if she approves it.

A question created by one of the reviewers is decided upon by the other two. If a question is created by Amal, then it is first reviewed by Bimal. If Bimal approves the question, then it is accepted. Otherwise, it is reviewed by Komal. The question is then accepted only if Komal approves it. A similar process is followed for questions created by Bimal, whose questions are first reviewed by Amal, and then by Amal only if Komal disapproves it. Questions created by Komal are first reviewed by Amal, and then, if required, by Bimal.

The following facts are known about the review process after its completion.



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Question 6

How many questions created by Amal or Bimal were disapproved by at least one of the other reviewers?

- A 7
- B 4
- C 5
- D 2

not reviewed by Komal. If both disapprove the question, it is rejected and is not reviewed by Komal. If one of them approves the question and the other disapproves it, then the question is reviewed by Komal. Then the question is accepted only if she approves it.

A question created by one of the reviewers is decided upon by the other two. If a question is created by Amal, then it is first reviewed by Bimal. If Bimal approves the question, then it is accepted. Otherwise, it is reviewed by Komal. The question is then accepted only if Komal approves it. A similar process is followed for questions created by Bimal, whose questions are first reviewed by Komal, and then by Amal only if Komal disapproves it. Questions created by Komal are first reviewed by Amal, and then, if required, by Bimal.

The following facts are known about the review process after its completion.

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Instructions

Venkat, a stockbroker, invested a part of his money in the stock of four companies — A, B, C and D. Each of these companies belonged to different industries, viz., Cement, Information Technology (IT), Auto, and Steel, in no particular order. At the time of investment, the price of each stock was Rs.100. Venkat purchased only one stock of each of these companies. He was expecting returns of 20%, 10%, 30%, and 40% from the stock of companies A, B, C and D, respectively. Returns are defined as the change in the value of the stock after one year, expressed as a percentage of the initial value. During the year, two of these companies announced extraordinarily good results. One of these two companies belonged to the Cement or the IT industry, while the other one belonged to either the Steel or the Auto industry. As a result, the returns on the stocks of these two companies were higher than the initially expected returns. For the company belonging to the Cement or the IT industry with extraordinarily good results, the returns were twice that of the initially expected returns. For the company belonging to the Steel or the Auto industry, the returns on announcement of extraordinarily good results were only one and a half times that of the initially expected returns. For the remaining two companies, which did not announce extraordinarily good results, the returns realized during the year were the same as initially expected.

Question 7

If Company C belonged to the Cement or the IT industry and did announce extraordinarily good results, then which of these statement(s) would necessarily be true?

- I. Venkat earned not more than 36.25% return on average.
- II. Venkat earned not less than 33.75% return on average.
- III. If Venkat earned 33.75% return on average, Company A announced extraordinarily good results.
- IV. If Venkat earned 33.75% return on average, Company B belonged either to Auto or to Steel Industry.

- A I and II only
- B II and IV only
- C II and III only
- D III and IV only

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Question 8

If Venkat earned a 38.75% return on average during the year, then which of these statement(s) would necessarily be true?

- I. Company C belonged either to Auto or to Steel Industry.
- II. Company D belonged either to Auto or to Steel Industry.
- III. Company A announced extraordinarily good results.
- IV. Company B did not announce extraordinarily good results.

- A I and II only
- B II and III only
- C I and IV only
- D II and IV only

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Question 9

What is the minimum average return Venkat would have earned during the year?

- A 30%
- B 31.25%
- C 32.5%
- D Cannot be determined

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Question 10

If Venkat earned a 35% return on average during the year, then which of these statements would necessarily be true?

- I. Company A belonged either to Auto or to Steel Industry.
- II. Company B did not announce extraordinarily good results.
- III. Company A announced extraordinarily good results.
- IV. Company D did not announce extraordinarily good results.

- A I and II only
- B II and III only
- C III and IV only
- D II and IV only

Instructions

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

Colour	Rs / liter
RED	20
YELLOW	25
WHITE	15
ORANGE	22
PINK	18

Rang Barsey Paint Company (RBPC) is in the business of manufacturing paints. RBPC buys RED, YELLOW, WHITE, ORANGE, and PINK paints. ORANGE paint can be also produced by mixing RED and YELLOW paints in equal proportions. Similarly, PINK paint can also be produced by mixing equal amounts of RED and WHITE paints. Among other paints, RBPC sells CREAM paint, (formed by mixing WHITE and YELLOW in the ratio 70:30) AVOCADO paint (formed by mixing equal amounts of ORANGE and PINK paint) and WASHEDORANGE paint (formed by mixing equal amounts of ORANGE and WHITE paint). The following table provides the price at which RBPC buys paints.

Question 11

Assume that AVOCADO, CREAM and WASHEDORANGE each sells for the same price. Which of the three is the most profitable to manufacture?

- A AVOCADO
- B CREAM
- C WASHEDORANGE
- D Sufficient data is not available.

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

Colour	Rs / liter
RED	20
YELLOW	25
WHITE	15
ORANGE	22
PINK	18

Rang Barsey Paint Company (RBPC) is in the business of manufacturing paints. RBPC buys RED, YELLOW, WHITE, ORANGE, and PINK paints. ORANGE paint can be also produced by mixing RED and YELLOW paints in equal proportions. Similarly, PINK paint can also be produced by mixing equal amounts of RED and WHITE paints. Among other paints, RBPC sells CREAM paint, (formed by mixing WHITE and YELLOW in the ratio 70:30) AVOCADO paint (formed by mixing equal amounts of ORANGE and PINK paint) and WASHEDORANGE paint (formed by mixing equal amounts of ORANGE and WHITE paint). The following table provides the price at which RBPC buys paints.

The cheapest way to manufacture AVOCADO paint would cost

A) Rs. 19.50 per litre.
B) Rs. 19.75 per litre.
C) Rs. 20.00 per litre.

Handwritten calculations:

- Orange = 22.5
- Pink = 17.5
- Cream = 18
- Avocado = 19.5

Example calculation for Orange: $15 \times 0.5 + 25 \times 0.5 = 20$

Question 12

The cheapest way to manufacture AVOCADO paint would cost

- A Rs. 19.50 per litre.
- B Rs. 19.75 per litre
- C Rs. 20.00 per litre.

D Rs. 20.25 per litre.

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

Colour	Rs / litre
RED	20
YELLOW	25
WHITE	15
ORANGE	22
PINK	18

Rang Barsey Paint Company (RBPC) is in the business of manufacturing RED, YELLOW, WHITE, ORANGE, and PINK paints. ORANGE paint can be also produced by mixing RED and YELLOW paints in equal proportions. Similarly, PINK paint can be produced by mixing equal amounts of RED and WHITE paints. Among other paints, RBPC sells CREAM paint, (formed by mixing WHITE and YELLOW in ratio 70:30) AVOCADO paint (formed by mixing equal amounts of ORANGE and PINK paint) and WASHEDORANGE paint (formed by mixing equal amounts of ORANGE and WHITE paint). The following table provides the price at which RBPC buys paints.

The cheapest way to manufacture AVOCADO paint will cost

A) Rs. 19.50 per litre.
B) Rs. 20.00 per litre.
C) Rs. 20.25 per litre.
D) Rs. 20.50 per litre.

Handwritten notes: $15 \times 0.7 + 25 \times 0.3 = 10.5 + 7.5 = 18$ (for CREAM), Orange - 22.5, Pink - 17.5, Cream - 18.



VIDEO SOLUTION



CAT Syllabus PDF



Question 13

WASHEDORANGE can be manufactured by mixing

- A CREAM and RED in the ratio 14:10.
- B CREAM and RED in the ratio 3:1.
- C YELLOW and PINK in the ratio 1:1.
- D RED, YELLOW, and WHITE in the ratio 1:1:2.

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

Colour	Rs / litre
RED	20
YELLOW	25
WHITE	15
ORANGE	22
PINK	18

Rang Barsey Paint Company (RBPC) is in the business of manufacturing RED, YELLOW, WHITE, ORANGE, and PINK paints. ORANGE paint can be also produced by mixing RED and YELLOW paints in equal proportions. Similarly, PINK paint can be produced by mixing equal amounts of RED and WHITE paints. Among other paints, RBPC sells CREAM paint, (formed by mixing WHITE and YELLOW in ratio 70:30) AVOCADO paint (formed by mixing equal amounts of ORANGE and PINK paint) and WASHEDORANGE paint (formed by mixing equal amounts of ORANGE and WHITE paint). The following table provides the price at which RBPC buys paints.

The cheapest way to manufacture AVOCADO paint will cost

A) Rs. 19.50 per litre.
B) Rs. 20.00 per litre.
C) Rs. 20.25 per litre.
D) Rs. 20.50 per litre.

Handwritten notes: $15 \times 0.7 + 25 \times 0.3 = 10.5 + 7.5 = 18$ (for CREAM), Orange - 22.5, Pink - 17.5, Cream - 18.



VIDEO SOLUTION

Instructions

For the following questions answer them individually

Question 14

Four friends Ashok, Bashir, Chirag and Deepak are out for shopping. Ashok has less money than three times the amount that Bashir has. Chirag has more money than Bashir. Deepak has an amount equal to the difference of amounts with Bashir and Chirag. Ashok has three times the money with Deepak. They each have to buy at least one shirt, or one shawl, or one sweater, or one jacket that are priced Rs. 200, Rs. 400, Rs. 600, and Rs. 1,000 a

piece respectively. Chirag borrows Rs. 300 from Ashok and buys a jacket. Bashir buys a sweater after borrowing Rs. 100 from Ashok and is left with no money. Ashok buys three shirts. What is the costliest item that Deepak could buy with his own money?

- A A shirt
- B A shawl
- C A sweater
- D A jacket

[VIEW SOLUTION](#)

Question 15

Eighty kilograms (kg) of store material is to be transported to a location 10 km away. Any number of couriers can be used to transport the material. The material can be packed in any number of units of 10,20 or 40kg. Couriers charges are Rs. 10 per hour. Couriers travel at the speed of 5 km/hr if carrying 10kg, at 2 km/hr if carrying 20kg and at 1 km/hr if carrying 40 kg. A courier cannot carry more than 40 kg of load. The minimum cost at which 80kg of store material can be transported to its destination will be:

- A Rs.180
- B Rs.160
- C Rs.140
- D Rs.120

[VIEW SOLUTION](#)



Question 16

I brought 30 books on Mathematics, Physics, and Chemistry, priced at Rs.17, Rs.19, and Rs.23 per book respectively, for distribution among poor students of Standard X of a school. The physics books were more in number than the Mathematics books but less than the Chemistry books, the difference being more than one. The total cost amounted to Rs.620. How many books on Mathematics, Physics, and Chemistry could have been bought respectively?

- A 5, 8, 17
- B 5, 12, 13
- C 5, 10, 15
- D 5, 6, 19

[VIEW SOLUTION](#)

Question 17

A king has unflinching loyalty from eight of his ministers M1 to M8, but he has to select only four to make a cabinet committee. He decides to choose these four such that each selected person shares a liking with at least one of the other three selected. The selected persons must also hate at least one of the likings of any of the other three persons selected.

M1 likes fishing and smoking, but hates gambling.

M2 likes smoking and drinking, but hates fishing.

M3 likes gambling, but hates smoking,

M4 likes mountaineering, but hates drinking,

M5 likes drinking, but hates smoking and mountaineering.

M6 likes fishing, but hates smoking and mountaineering.

M7 likes gambling and mountaineering, but hates fishing.

M8 likes smoking and gambling, but hates mountaineering.

Who are the four people selected by the king?

- A M1, M2, M5 and M6
- B M3, M4, M5 and M6
- C M4, M5, M6 and M8
- D M1, M2, M4 and M7

 [VIEW SOLUTION](#)

Question 18

My bag can carry no more than ten books, I must carry at least one book each of management, mathematics, physics and fiction. Also, for every management book I carry I must carry two or more fiction books, and for every mathematics book I carry I must carry two or more physics books. I earn 4, 3, 2 and 1 points for each management, mathematics, physics and fiction book, respectively, I carry in my bag. I want to maximise the points I can earn by carrying the most appropriate combination of books in my bag. The maximum points that I can earn are:

- A 20
- B 21
- C 22
- D 23

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Question 19

There are ten animals — two each of lions, panthers, bison, bears, and deer — in a zoo. The enclosures in the zoo are named X, Y, Z, P and Q and each enclosure is allotted to one of the following attendants: Jack, Mohan, Shalini, Suman and Rita. Two animals of different species are housed in each enclosure. A lion and a deer cannot be together. A panther cannot be with either a deer or a bison. Suman attends to animals from among bison, deer, bear and panther only. Mohan attends to a lion and a panther. Jack does not attend to deer, lion or bison. X, Y and Z are allotted to Mohan, Jack and Rita respectively. X and Q enclosures have one animal of the same species. Z and P have the same pair of animals. The animals attended by Shalini are:

- A bear & bison
- B bison & deer
- C bear & lion
- D bear & panther

[VIEW SOLUTION](#)

Question 20

I have a total of Rs. 1,000. Item A costs Rs. 110, item B costs Rs. 90, item C costs Rs. 70, item D costs Rs. 40 and item E costs Rs. 45. For every item D that I purchase, I must also buy two of item B. For every item A, I must buy one of item C. For every item E, I must also buy two of item D and one of item B. For every item purchased I earn 1,000 points and for every rupee not spent I earn a penalty of 1,500 points. My objective is to maximise the points I earn. What is the number of items that I must purchase to maximise my points?

- A 13
- B 14
- C 15
- D 16

[VIEW SOLUTION](#)

Instructions

K, L, M, N, P, Q, R, S, U and W are the only ten members in a department. There is a proposal to form a team from within the members of the department, subject to the following conditions:

1. A team must include exactly one among P, R and S.
2. A team must include either M or Q, but not both.
3. If a team includes K, then it must also include L, and vice versa.
4. If a team includes one among S, U and W, then it should also include the other two.
5. L and N cannot be members of the same team.
6. L and U cannot be members of the same team.

The size of a team is defined as the number of members in the team.

Question 21

What could be the size of a team that includes K?

- A 2 or 3
- B 2 or 4
- C 3 or 4
- D Only 2
- E Only 4

K, L, M, N, P, Q, R, S, U and W are the only ten members in a department. There is a proposal to form a team from within the members of the department, subject to the following conditions:

1. A team must include exactly one among P, R and S.
2. A team must include either Q or U, but not both.
3. If a team includes K, then it must also include L, and vice versa.
4. If a team includes one among S, U and W, then it should also include the other two.
5. L and N cannot be members of the same team.
6. L and U cannot be members of the same team.

The size of a team is defined as the number of members in the team.

Handwritten notes on the screen show a diagram with three columns labeled P, R, and S. Under P, there is a cross and 'R, S' and 'U, W'. Under R, there is a cross and 'P, S' and 'U, W'. Under S, there is a cross and 'U, W' and 'P, R'. A large play button is overlaid on the screen.

VIDEO SOLUTION

CAT Percentile Predictor



Question 22

Who cannot be a member of a team of size 3?

- A L
- B M
- C N
- D P
- E Q

K, L, M, N, P, Q, R, S, U and W are the only ten members in a department. There is a proposal to form a team from within the members of the department, subject to the following conditions:

1. A team must include exactly one among P, R and S.
2. A team must include either M or Q, but not both.
3. If a team includes K, then it must also include L, and vice versa.
4. If a team includes one among S, U and W, then it should also include the other two.
5. L and N cannot be members of the same team.
6. L and U cannot be members of the same team.

The size of a team is defined as the number of members in the team.

Handwritten notes on the screen show a diagram with three columns labeled P, R, and S. Under P, there is a cross and 'R, S' and 'U, W'. Under R, there is a cross and 'P, S' and 'U, W'. Under S, there is a cross and 'U, W' and 'P, R'. A large play button is overlaid on the screen.

VIDEO SOLUTION

Question 23

Who can be a member of a team of size 5?

- A K
- B L
- C M

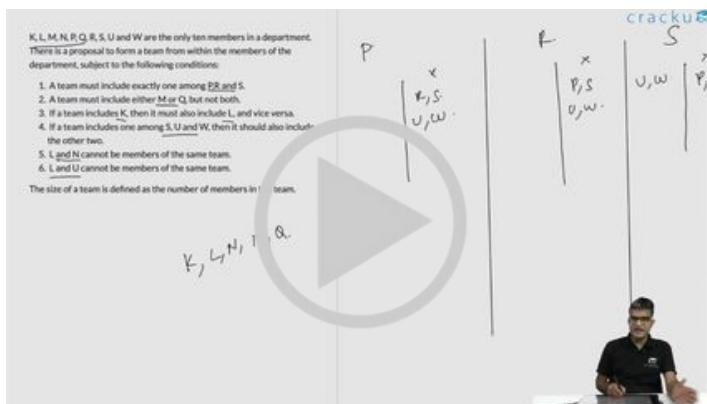
D P

E R

K, L, M, N, P, Q, R, S, U and W are the only ten members in a department. There is a proposal to form a team from within the members of the department, subject to the following conditions:

1. A team must include exactly one among P, R and S.
2. A team must include either M or Q, but not both.
3. If a team includes K, then it must also include L, and vice versa.
4. If a team includes one among S, U and W, then it should also include the other two.
5. L and N cannot be members of the same team.
6. L and U cannot be members of the same team.

The size of a team is defined as the number of members in the team.



VIDEO SOLUTION

Question 24

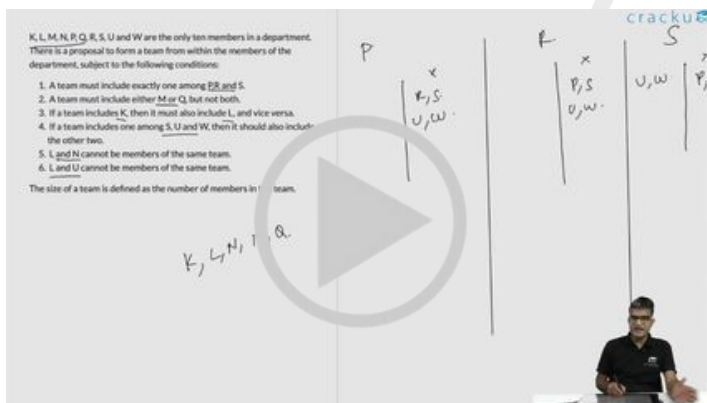
In how many ways a team can be constituted so that the team includes N?

- A 2
- B 3
- C 4
- D 5
- E 6

K, L, M, N, P, Q, R, S, U and W are the only ten members in a department. There is a proposal to form a team from within the members of the department, subject to the following conditions:

1. A team must include exactly one among P, R and S.
2. A team must include either M or Q, but not both.
3. If a team includes K, then it must also include L, and vice versa.
4. If a team includes one among S, U and W, then it should also include the other two.
5. L and N cannot be members of the same team.
6. L and U cannot be members of the same team.

The size of a team is defined as the number of members in the team.



VIDEO SOLUTION



CAT Score Calculator

Question 25

What would be the size of the largest possible team?

- A 8

- E** cannot be determined

 VIDEO SOLUTION

Directions for the following three questions: Answer the questions based on the passage below.

Question 26

D Fiza, David, Ram and Peter

Directions for the following three questions: Answer the questions based on the passage below.

A group of three or four has to be selected from seven persons. Among the seven are two women: Fiza and Kavita, and five men: Ram, Shyam, David, Peter and Rahim. Ram would not like to be in the group if Shyam is also selected. Shyam and Rahim want to be selected together in the group. Kavita would like to be in the group only if David is also there. David, if selected, would not like Peter in the group. Ram would like to be in the group only if Peter is also there. David insists that Fiza be selected in case he is there in the group.

Which of the following is a feasible group of three?

- A) David, Ram and Rahim
C) Kavita, David and Shyam

- D) Fiza, Dayal and Ram

Ram, Shyam, David,
Peter,
Rahim Fiza, Kavita

Question 27

Which of the following is a feasible group of three?

- A David, Ram and Rahim
- B Peter, Shyam and Rahim
- C Kavita, David and Shyam
- D Fiza, David and Ram

Directions for the following three questions: Answer the questions based on the passage below.

A group of three or four has to be selected from seven persons. Among the seven are two women: Fiza and Kavita, and five men: Ram, Shyam, David, Peter and Rahim. Ram would not like to be in the group if Shyam is also selected. Shyam and Rahim want to be selected together in the group. Kavita would like to be in the group only if David is also there. David, if selected, would not like Peter in the group. Ram would like to be in the group only if Peter is also there. David insists that Fiza be selected in case he is there in the group.

Which of the following is a feasible group of three?

A) David, Ram and Rahim
B) Peter, Shyam and Rahim
C) Kavita, David and Shyam
D) Fiza, David and Ram

Handwritten notes: Ram, Shyam, David, Peter, Fiza, Kavita.

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Question 28

Which of the following statements is true?

- A Kavita and Ram can be part of a group of four
- B A group of four can have two women
- C A group of four can have all four men
- D None of these

Directions for the following three questions: Answer the questions based on the passage below.

A group of three or four has to be selected from seven persons. Among the seven are two women: Fiza and Kavita, and five men: Ram, Shyam, David, Peter and Rahim. Ram would not like to be in the group if Shyam is also selected. Shyam and Rahim want to be selected together in the group. Kavita would like to be in the group only if David is also there. David, if selected, would not like Peter in the group. Ram would like to be in the group only if Peter is also there. David insists that Fiza be selected in case he is there in the group.

Which of the following is a feasible group of three?

A) David, Ram and Rahim
B) Peter, Shyam and Rahim
C) Kavita, David and Shyam
D) Fiza, David and Ram

Handwritten notes: Ram, Shyam, David, Peter, Fiza, Kavita.

Instructions

Two traders, Chetan and Michael, were involved in the buying and selling Of MCS shares over five trading days. At the beginning of the first day, the MCS share was priced at Rs 100, while at the end of the fifth day it was priced at Rs 110. At the end of each day, the MCS share price either went up by Rs 10, or else, it came down by Rs 10. Both Chetan and Michael took buying and selling decisions at the end of each trading day. The beginning price of MCS share on a given day was the same as the ending price of the previous day. Chetan and Michael started with the same number of shares and amount of cash, and had enough of both. Below are some additional facts about how Chetan and Michael traded over the five trading days.

- Each day if the price went up, Chetan sold 10 shares of MCS at the closing price. On the other hand, each day if the price went down, he bought 10 shares at the closing price.
- If on any day, the closing price was above Rs 110, then Michael sold 10 shares of MCS, while if it was below Rs 90, he bought 10 shares, all at the closing price.

Question 29

If Chetan ended up with Rs 1300 more cash than Michael at the end of day 5, what was the price of MCS share at the end of day 4?

- A Rs 90
- B Rs 100
- C Rs 110
- D Rs 120
- E Not uniquely determinable

	1	2	3	4	5
1	90	80	90	100	110
2	90	100	90	100	110
3	90	100	110	100	110
4	90	100	110	120	110
5	110	100	90	100	110
6	110	100	110	100	110
7	110	100	110	120	110
8	110	120	110	100	110
9	110	120	110	120	110
10	110	120	130	120	110

decisions at the end of each trading day. The beginning price of MCS share on a given day was the same as the ending price of the previous day. Chetan and Michael started with the same number of shares and amount of cash, and had enough of both. Below are some additional facts about how Chetan and Michael traded over the five trading days.

- Each day if the price went up, Chetan sold 10 shares of MCS at the closing price. On the other hand, each day if the price went down, he bought 10 shares at the closing price.
- If on any day, the closing price was above Rs 110, then Michael sold 10 shares of MCS, while if it was below Rs 90, he bought 10 shares, all at the closing price.

Question 30

What could have been the maximum possible increase in combined cash balance of Chetan and Michael at the end of the fifth day?

- A Rs 3700
- B Rs 4000
- C Rs 4700

D Rs 5000

E Rs 6000

	1	2	3	4	5
1	90	80	90	100	110
2	90	100	90	100	110
3	90	100	110	100	110
4	90	100	110	120	110
5	110	100	90	100	110
6	110	100	110	100	110
7	110	100	110	120	110
8	110	120	110	100	110
9	110	120	110	120	110
10	110	120	130	120	110

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- On the other hand, each day if the price went down, he bought 10 shares at the closing price.
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VIDEO SOLUTION





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Question 31

If Chetan sold 10 shares of MCS on three consecutive days, while Michael sold 10 shares only once during the five days, what was the price of MCS at the end of day 3?

- A Rs 90
- B Rs 100
- C Rs 110
- D Rs 120
- E Rs 130

	1	2	3	4	5
1	90	80	90	100	110
2	90	100	90	100	110
3	90	100	110	100	110
4	90	100	110	120	110
5	110	100	90	100	110
6	110	100	110	100	110
7	110	100	110	120	110
8	110	120	110	100	110
9	110	120	110	120	110
10	110	120	130	120	110

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- Each day if the price went up, Chetan sold 10 shares of MCS at the closing price.
- On the other hand, each day if the price went down, he bought 10 shares at the closing price.
- If on any day, the closing price was above Rs 110, then Michael sold 10 shares of MCS, while if it was below Rs 90, he bought 10 shares, all at the closing price.

VIDEO SOLUTION

Question 32

If Michael ended up with Rs 100 less cash than Chetan at the end of day 5, what was the difference in the number of shares possessed by Michael and Chetan (at the end of day 5)?

- A Michael had 10 less shares than Chetan.
- B Michael had 10 more shares than Chetan.
- C Chetan had 10 more shares than Michael.
- D Chetan had 20 more shares than Michael.
- E Both had the same number of shares.

	1	2	3	4	5
1	90	80	90	100	110
2	90	100	90	100	110
3	90	100	110	100	110
4	90	100	110	120	110
5	110	100	90	100	110
6	110	100	110	100	110
7	110	100	110	120	110
8	110	120	110	100	110
9	110	120	110	120	110
10	110	120	130	120	110

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- Each day if the price went up, Chetan sold 10 shares of MCS at the closing price.
- On the other hand, each day if the price went down, he bought 10 shares at the closing price.
- If on any day, the closing price was above Rs 110, then Michael sold 10 shares of MCS, while if it was below Rs 90, he bought 10 shares, all at the closing price.

VIDEO SOLUTION

Question 33

If Michael ended up with 20 more shares than Chetan at the end of day 5, what was the price of the share at the end of day 3?

- A Rs 90
- B Rs 100
- C Rs 110
- D Rs 120
- E Rs 130

	1	2	3	4	5
1	90	80	90	100	110
2	90	100	90	100	110
3	90	100	110	100	110
4	90	100	110	120	110
5	110	100	90	100	110
6	110	100	110	100	110
7	110	100	110	120	110
8	110	120	110	100	110
9	110	120	110	120	110
10	110	120	130	120	110

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- If on any day, the closing price was above Rs 110, then Michael sold 10 shares of MCS, while if it was below Rs 90, he bought 10 shares, all at the closing price.

VIDEO SOLUTION

Instructions

Directions for the following four questions: Answer the questions on the basis of the information given below.

Twenty one participants from four continents (Africa, Americas, Australasia, and Europe) attended a United Nations conference. Each participant was an expert in one of four fields, labour, health, population studies, and refugee relocation. The following five facts about the participants are given.

- (a) The number of labour experts in the camp was exactly half the number of experts in each of the three other categories.
- (b) Africa did not send any labour expert. Otherwise, every continent, including Africa, sent at least one expert for each category.
- (c) None of the continents sent more than three experts in any category.
- (d) If there had been one less Australasian expert, then the Americas would have had twice as many experts as each of the other continents.
- (e) Mike and Alfanzo are leading experts of population studies who attended the conference. They are from Australasia.

Question 34

If Ramos is the lone American expert in population studies, which of the following is NOT true about the numbers of experts in the conference from the four continents?

- A There is one expert in health from Africa.
- B There is one expert in refugee relocation from Africa.
- C There are two experts in health from the Americas.
- D There are three experts in refugee relocation from the Americas.



 VIDEO SOLUTION

Question 35

Which of the following combinations is NOT possible?

- A 2 experts in population studies from the Americas and 2 health experts from Africa attended the conference.

- B** 2 experts in population studies from the Americas and 1 health expert from Africa attended the conference.
- C** 3 experts in refugee relocation from the Americas and 1 health expert from Africa attended the conference.
- D** Africa and America each had 1 expert in population studies attending the conference.



[VIDEO SOLUTION](#)

Question 36

Alex, an American expert in refugee relocation, was the first keynote speaker in the conference. What can be inferred about the number of American experts in refugee relocation in the conference, excluding Alex?

- i. At least one
- ii. At most two
- A** Only i and not ii
- B** Only ii and not i
- C** Both i and ii
- D** Neither i nor ii



[VIDEO SOLUTION](#)

Question 37

Which of the following numbers cannot be determined from the information given?

- A Number of labour experts from the Americas.
- B Number of health experts from Europe.
- C Number of health experts from Australasia.
- D Number of experts in refugee relocation from Africa.



VIDEO SOLUTION

Instructions

In the table below the check marks indicate all languages spoken by five people: Paula, Quentin, Robert, Sally and Terence. For example, Paula speaks only Chinese and English.

	Arabic	Basque	Chinese	Dutch	English	French
Paula			✓		✓	
Quentin				✓	✓	
Robert	✓					✓
Sally		✓			✓	
Terence			✓			✓

These five people form three teams, Team 1, Team 2 and Team 3. Each team has either 2 or 3 members. A team is said to speak a particular language if at least one of its members speak that language.

The following facts are known.

- (1) Each team speaks exactly four languages and has the same number of members.
- (2) English and Chinese are spoken by all three teams, Basque and French by exactly two teams and the other languages by exactly one team.
- (3) None of the teams include both Quentin and Robert.
- (4) Paula and Sally are together in exactly two teams.
- (5) Robert is in Team 1 and Quentin is in Team 3.

Question 38

Who among the following four people is a part of exactly two teams?

- A Paula

- B Quentin
- C Sally
- D Robert

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In the table below the check marks indicate all languages spoken by five people: Paula, Quentin, Robert, Sally and Terence. For example, Paula speaks only Chinese and English.

	Arabic	Banque	Chinese	Dutch	English	French
Paula			✓		✓	
Quentin				✓		
Robert	✓					
Sally				✓	✓	✓
Terence						✓

These five people form three teams, Team 1, Team 2 and Team 3. Each team has either 2 or 3 members. A team is said to speak a particular language if at least one of its members speak that language. The following facts are known.

- (1) Each team speaks exactly four languages and has the same number of members.
- (2) English and Chinese are spoken by all three teams, but Arabic and French by exactly two teams and the other languages by exactly one team.
- (3) None of the teams include both Quentin and Robert.
- (4) Paula and Sally are together in exactly two teams.
- (5) Robert is in Team 1 and Quentin is in Team 3.

Handwritten notes on the right:

Languages

Team 1: Arabic, Chinese, English, French

Team 2: Arabic, Chinese, English, French

Team 3: Arabic, Chinese, English, French

Person

Team 1: Robert, Sally, Terence

Team 2: Paula, Quentin, Sally, Terence

Team 3: Paula, Quentin, Robert, Sally, Terence

VIDEO SOLUTION

Question 39

Who among the following four is not a member of Team 2?

- A Paula
- B Terence
- C Quentin
- D Sally

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In the table below the check marks indicate all languages spoken by five people: Paula, Quentin, Robert, Sally and Terence. For example, Paula speaks only Chinese and English.

	Arabic	Banque	Chinese	Dutch	English	French
Paula			✓		✓	
Quentin				✓		
Robert	✓					
Sally				✓	✓	✓
Terence						✓

These five people form three teams, Team 1, Team 2 and Team 3. Each team has either 2 or 3 members. A team is said to speak a particular language if at least one of its members speak that language. The following facts are known.

- (1) Each team speaks exactly four languages and has the same number of members.
- (2) English and Chinese are spoken by all three teams, but Arabic and French by exactly two teams and the other languages by exactly one team.
- (3) None of the teams include both Quentin and Robert.
- (4) Paula and Sally are together in exactly two teams.
- (5) Robert is in Team 1 and Quentin is in Team 3.

Handwritten notes on the right:

Languages

Team 1: Arabic, Chinese, English, French

Team 2: Arabic, Chinese, English, French

Team 3: Arabic, Chinese, English, French

Person

Team 1: Robert, Sally, Terence

Team 2: Paula, Quentin, Sally, Terence

Team 3: Paula, Quentin, Robert, Sally, Terence

VIDEO SOLUTION

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CAT Study Material

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Image of two students and a laptop.

Question 40

Who among the five people is a member of all teams?

- A Terence

- B Sally
- C Paula
- D No one

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In the table below the check marks indicate all languages spoken by five people: Paula, Quentin, Robert, Sally and Terence. For example, Paula speaks only Chinese and English.

	Arabic	Basque	Chinese	Dutch	English	French
Paula			✓		✓	
Quentin	✓					✓
Robert		✓			✓	
Sally	✓				✓	
Terence				✓		✓

These five people form three teams, Team 1, Team 2 and Team 3. Each team has either 2 or 3 members. A team is said to speak a particular language if at least one of its members speak that language. The following facts are known.

- (1) Each team speaks exactly four languages and has the same number of members.
- (2) English and Chinese are spoken by all three teams, but French by exactly two teams and the other languages by exactly one team.
- (3) None of the teams include both Quentin and Robert.
- (4) Paula and Sally are together in exactly two teams.
- (5) Robert is in Team 1 and Quentin is in Team 3.

Handwritten notes on the right side of the screen show a table for 'Languages' and 'Person' with rows for Team 1, Team 2, and Team 3, and columns for each language. A person is visible in the bottom right corner of the video frame.

VIDEO SOLUTION

Question 41

Apart from Chinese and English, which languages are spoken by Team 1?

- A Arabic and French
- B Basque and French
- C Arabic and Basque
- D Basque and Dutch

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In the table below the check marks indicate all languages spoken by five people: Paula, Quentin, Robert, Sally and Terence. For example, Paula speaks only Chinese and English.

	Arabic	Basque	Chinese	Dutch	English	French
Paula			✓		✓	
Quentin	✓					✓
Robert		✓			✓	
Sally	✓				✓	
Terence				✓		✓

These five people form three teams, Team 1, Team 2 and Team 3. Each team has either 2 or 3 members. A team is said to speak a particular language if at least one of its members speak that language. The following facts are known.

- (1) Each team speaks exactly four languages and has the same number of members.
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- (3) None of the teams include both Quentin and Robert.
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VIDEO SOLUTION